

Mahira Khan

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EDUCATION

Dr. Akhilesh Das Gupta Institute of Professional Studies (ADGIPS), GGSIPU Delhi, India
Bachelor of Technology in Information Technology; CGPA: 8.6/10 *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Discrete Mathematics, Operating Systems, Computer Networks, Computer Architecture, Design & Analysis of Algorithms

PROJECTS

SyncBoard | *Next.js, React, MongoDB, Socket.io, TanStack Query, Tailwind CSS* 2026

- Built a real-time collaborative Kanban board supporting concurrent multi-user editing via Socket.io, with live presence indicators and card-lock overlays to prevent conflicting edits
- Engineered optimistic concurrency control using MongoDB's `findOneAndUpdate` with version fields, eliminating race conditions when multiple users simultaneously moved the same card
- Implemented optimistic UI updates with TanStack Query's snapshot/revert pattern, giving users instant feedback while gracefully rolling back on server conflicts
- Designed a LexoRank-based ordering system using float positions for card reordering, reducing update complexity from $O(n)$ to $O(1)$ versus naive integer-based reordering
- Integrated GitHub/Google OAuth via NextAuth.js v5 with server-side protected routes, and built fully keyboard-accessible drag-and-drop (WCAG compliant) using @dnd-kit

VeloxDB – In-Memory Key-Value Store | *Multithreading, File I/O* 2026

- Designed and implemented an in-memory key-value store in Java inspired by Redis, supporting TTL-based expiry, LRU eviction, and thread-safe concurrent operations
- Built thread-safe read/write operations using `ConcurrentHashMap`, enabling safe concurrent access from multiple threads without external locking
- Implemented an LRU eviction policy using a `LinkedHashMap`-based cache to enforce configurable memory limits and evict least-recently-used entries
- Developed lazy expiry checks and a background cleanup thread to automatically purge expired keys, alongside file-based persistence for save/reload on restart
- Simulated concurrent multi-threaded access patterns to validate correctness of thread-safety guarantees under load

ACHIEVEMENTS

Top 22 of 2,500+ teams – Code For Bharat Season 2 (*Microsoft; Tech Masters India*) 2025

- Built CampusHustle, a freelance marketplace and skill-matching platform connecting students with real-world project opportunities

Best Girl Team, Top 10 of 1,700+ teams – Hackster 2025 (*HMR Institute of Technology*) 2025

- Built Astitva, an AI-powered life companion app unifying mental health, career, and personal growth tracking

Top 10 team – TechJam 2.0 (*Microsoft Sovereign Office, Noida; MLH, Unstop*) 2025

PROBLEM SOLVING & DSA

LeetCode: 180+ problems solved across C++, Java, and JavaScript; earned the 100 Days Consistency Badge (2026)

Core Strength: Arrays, Strings, LinkedList, Bit manipulation, Heaps, Hash Tables, Sliding Window

Currently Advancing: Dynamic Programming, Backtracking, Divide & Conquer, Greedy, Graphs